Collaborative Wind Tunnel Flight Investigation

Multi-Variable Functional Scaling of Aerodynamic Stall

$$\alpha_{\text{stall}} = f(\alpha,\, m,\, t/c,\, Re)$$

Platform: NCSSM *WindTunnel lab* (GPU-Accelerated 2D Lattice Boltzmann Simulator)

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Mission Briefing & Research Objective

In computational aerodynamics and experimental flight testing, **aerodynamic stall** represents the catastrophic loss of lift and surge in pressure drag resulting from massive boundary layer separation. While idealized thin-airfoil theory predicts an unyielding linear lift curve $C_L(\alpha) = 2\pi(\alpha - \alpha_0)$, physical fluids cannot overcome the adverse pressure gradient on the suction surface indefinitely.
As an aerospace research team, our class is commissioned to map the **functional response surface** of the critical stall angle:

$$\alpha_{\text{stall}} = \Phi\left(m,\, \frac{t}{c},\, Re\right), \quad \text{and the maximum achievable lift} \quad C_{L,\max} = \Psi\left(m,\, \frac{t}{c},\, Re\right) \tag{1}$$

where m is maximum airfoil camber, t/c is the maximum thickness ratio, and Re is the chord Reynolds number. To accomplish this broad parametric exploration, the class will operate as an **Aerodynamics Research Consortium** divided into four specialized test divisions. Each division will interrogate one parametric axis, share verified digital flight telemetry, and collaborate on a unified global regression model.

1 Aerodynamic Theory & Physical Foundations

1.1 Lifting Airfoil Kinematics and Classical Limits

According to classical Kutta-Joukowski lift theory and 2D thin-airfoil theory, for an inviscid, attached flow around a slender foil with mean camber line $y_c(x)$, the sectional lift coefficient is linear:

$$C_L(\alpha) = 2\pi(\alpha - \alpha_0) = a_0(\alpha - \alpha_0), \quad a_0 \approx 2\pi \approx 6.28\,\text{rad}^{-1} \approx 0.110\,\text{deg}^{-1}, \tag{2}$$

where α_0 represents the zero-lift angle of attack. For NACA 4-digit airfoils parameterized by maximum camber m (expressed as a fraction of chord) located at chordwise station p, thin airfoil theory yields the integral:

$$\alpha_0 = -\frac{1}{\pi}\int_0^{\pi} \frac{dy_c}{dx}(\cos\theta - 1)\, d\theta \approx -k_c \cdot m, \quad (k_c \approx 1.05 \text{ to } 1.15 \text{ rad/unit camber}). \tag{3}$$

Hence, camber translates the lift curve to the left, generating positive lift at $\alpha = 0°$.

1.2 Adverse Pressure Gradients and Boundary Layer Separation

In a real viscous fluid, fluid deceleration downstream of the suction peak generates an adverse pressure gradient:

$$\frac{dp}{dx} > 0 \implies \mu \frac{\partial^2 u}{\partial y^2} > 0 \quad \text{at the wall } (y = 0). \tag{4}$$

Because viscous dissipation continuously strips kinetic energy from the near-wall fluid, the wall shear stress decreases until:

$$\tau_w = \mu \frac{\partial u}{\partial y}\bigg|_{y=0} = 0 \quad \implies \quad \text{Flow Separation Onset Point } x_{\text{sep}}. \tag{5}$$

Downstream of x_{sep}, backflow develops, shedding shear layers detach into the freestream, and an unsteady wake forms, culminating in aerodynamic stall.

1.3 Governing Factors in Stall Mechanics

1. **Airfoil Thickness Ratio (t/c):** Controls the leading-edge nose radius:

$$r_{\text{le}} \approx 1.1019 \left(\frac{t}{c}\right)^2 c. \tag{6}$$

   Thin airfoils ($t/c \leq 0.08$) exhibit sharp suction peaks and suffer *leading-edge thin-airfoil stall* with abrupt vortex bursting. Moderate-to-thick airfoils ($t/c \geq 0.12$) feature rounded leading edges that alleviate suction spikes, promoting progressive *trailing-edge stall* that delays α_{stall} to higher angles.

2. **Airfoil Camber (m):** Asymmetry increases suction surface curvature, shifting the stagnation point to the lower surface. While this amplifies bound circulation Γ and increases C_L at a given α, the intensified suction peak exacerbates dp/dx, which can trigger premature boundary layer separation if thickness is insufficient.

3. **Reynolds Number ($Re = U_\infty c/\nu$):** Governs the ratio of inertial transport to viscous dissipation. The laminar boundary layer thickness scales as:

$$\delta(x) \sim \frac{x}{\sqrt{Re_x}} \propto \frac{c}{\sqrt{Re}}. \tag{7}$$

   Higher Re imparts greater near-wall momentum, thinning the boundary layer and enabling it to climb steeper adverse pressure hills without detaching, thereby increasing α_{stall}.

2 Experimental Testing Platform: NCSSM *WindTunnel lab*

Tests are conducted using NCSSM's **WindTunnel lab**, a real-time GPU-accelerated numerical wind tunnel powered by the Lattice Boltzmann Method (LBM D2Q9) and Smagorinsky subgrid-scale turbulence modeling.

2.1 Standardized Data Acquisition (DAQ) Protocol

To ensure reproducibility across all student research teams, each data point must be collected according to the **5-Second Integer Vortex Shedding Integration Protocol**:

1. **Geometry Configuration:** Enter the 4-digit NACA profile (e.g., `2412`) in the control panel.

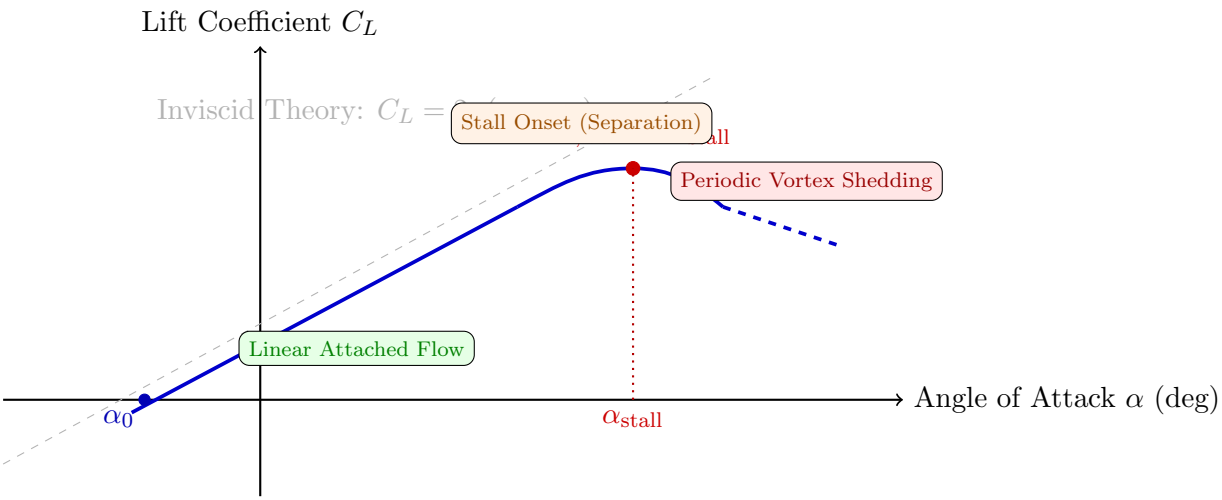


Figure 1: Anatomy of an aerodynamic lift polar: Linear attached regime, non-linear separation onset, peak lift $C_{L,\max}$ at critical stall angle α_{stall}, and post-stall periodic vortex shedding.

2. **Flow Conditioning:** Set the target Reynolds number (Re) slider and verify that *Smagorinsky LES* is checked.

3. **Angle Dialing:** Adjust α using the slider for coarse sweeping (2.0° increments) and the stepper buttons (`[<]` / `[>]`) for high-resolution refinement (0.5° increments) in the vicinity of stall.

4. **Flow Settling Verification:** Observe the telemetry status indicator. When flow is altered, the indicator displays **Flow Settling**. Wait 3–5 seconds until the badge turns **Flow Settled · Ready**.

5. **Execute 5s Cycle Acquisition:** Click `[Acquire (5s)]` or press hotkey `L`. The system integrates instantaneous momentum-exchange forces across full shedding periods ($\int \sin(\omega t)dt \equiv 0$), completely filtering phase bias.

6. **Telemetry Logging:** The polar graph displays the cycle-averaged C_L, accompanied by dynamic error whiskers representing the shedding envelope $[C_{L,\min}, C_{L,\max}]$.

7. **CSV Export:** Upon completing an angle sweep, click `[CSV]` to save the raw laboratory dataset.

3 Consortium Division Assignments

The class is partitioned into four research divisions. Each division is responsible for executing high-density angle of attack sweeps across their assigned parameter matrix.

3.1 Division 1: Airfoil Thickness (t/c) & Nose Curvature Division

Primary Objective: Quantify the impact of leading-edge radius and airfoil thickness on stall onset angle and determine whether thicker profiles experience sudden or docile stall.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Camber $m = 0\%$ (Symmetric Series).
- **Airfoil Sweep Matrix:**
  1. **NACA 0006:** Ultra-thin ($t/c = 6\%$) — Sharp leading edge, high susceptibility to separation bubble burst.
  2. **NACA 0009:** Moderately thin ($t/c = 9\%$) — Classic high-speed profile.
  3. **NACA 0012:** Standard benchmark ($t/c = 12\%$) — NASA calibration standard.
  4. **NACA 0018:** Thick subsonic ($t/c = 18\%$) — High structural depth, blunt nose.
  5. **NACA 0024:** Very thick ($t/c = 24\%$) — Low-speed strut/root section.
- **Angle Sweep Range:** $\alpha \in [-2.0°, +24.0°]$. Sweep in 2.0° steps from $-2°$ to 8°, then 0.5° steps from 8° to 20°.
- **Deliverable Plot 1.1:** Composite polar plot (C_L vs α) overlaying all five thickness ratios.
- **Deliverable Plot 1.2:** Functional curve of α_{stall} vs (t/c) and $C_{L,\max}$ vs (t/c).

3.2 Division 2: Airfoil Camber (m) & Zero-Lift Asymmetry Division

Primary Objective: Determine how mean camber line curvature shifts the zero-lift angle α_0, alters maximum lift $C_{L,\max}$, and affects the stall angle α_{stall}.

- **Fixed Controls:** Baseline Reynolds number $Re = 1000$, Thickness $t/c = 12\%$, Maximum Camber Location $p = 0.4c$ (`NACA X412` Family).
- **Airfoil Sweep Matrix:**
  1. **NACA 0012:** Symmetric reference ($m = 0\%$).
  2. **NACA 1412:** Mild camber ($m = 1\%$).
  3. **NACA 2412:** Benchmark camber ($m = 2\%$, typical general aviation wing).
  4. **NACA 4412:** High-lift camber ($m = 4\%$).
  5. **NACA 6412:** Extreme aggressive camber ($m = 6\%$, STOL/cargo section).
- **Angle Sweep Range:** $\alpha \in [-8.0°, +22.0°]$. Sweep in 2.0° steps from $-8°$ to 4°, then 0.5° steps through stall.
- **Deliverable Plot 2.1:** Experimental lift slope verification: verify linear slope $dC_L/d\alpha$ against 2π and plot experimental α_0 vs theoretical prediction $\alpha_0 \approx -1.1m$.
- **Deliverable Plot 2.2:** Functional curve of α_{stall} vs m and $C_{L,\max}$ vs m. Does increasing camber delay or advance stall onset?

3.3 Division 3: Reynolds Number (Re) & Viscous Scale Division

Primary Objective: Determine the empirical scaling law connecting Reynolds number to boundary layer energization and stall angle delay.

- **Fixed Controls:** Standard NACA 2412 airfoil profile ($m = 2\%$, $p = 0.4c$, $t/c = 12\%$).
- **Reynolds Number Sweep Matrix:**
  1. $Re = 300$**:** Viscous-dominated flow (thick laminar boundary layer, micro-aerial scale).
  2. $Re = 600$**:** Low Reynolds number transitional regime.
  3. $Re = 1200$**:** Standard baseline tunnel scale.
  4. $Re = 2500$**:** Moderate Reynolds number with elevated momentum transport.
  5. $Re = 5000$**:** High Reynolds number testing (energized boundary layer).
- **Angle Sweep Range:** $\alpha \in [-4.0^\circ, +22.0^\circ]$ for each Reynolds number. Focus 0.5° resolution in the $[8^\circ, 18^\circ]$ stall zone.
- **Deliverable Plot 3.1:** Family of $C_L(\alpha)$ curves showing the progression of stall softening and delay with increasing Re.
- **Deliverable Plot 3.2:** Semilogarithmic scaling plot: α_{stall} vs $\ln(Re)$ and test for power-law correlation $\alpha_{\text{stall}} \propto Re^k$.

3.4 Division 4: Coupled Interaction & Post-Stall Shedding Dynamics Division

Primary Objective: Investigate coupled camber-thickness interactions and quantify dynamic vortex shedding oscillation amplitudes and inverted stall asymmetry.

- **Sweep Matrix A (Coupled Geometry Matrix):**
  1. **NACA 4406:** High camber (4%) + Ultra-thin (6%) — Severe suction peak combined with sharp nose.
  2. **NACA 4418:** High camber (4%) + Thick (18%) — Camber-thickness synergy.
  3. **NACA 0006 vs NACA 0018:** Thickness extremes under symmetric conditions.
- **Sweep Matrix B (Bidirectional Inverted Stall Exploration):**
  1. Execute a continuous bidirectional sweep on **NACA 2412** from $\alpha = -16.0^\circ$ to $+20.0^\circ$.
  2. Locate positive stall $(\alpha^+_{\text{stall}}, C^+_{L,\max})$ and inverted negative stall $(\alpha^-_{\text{stall}}, C^-_{L,\min})$.
- **Deliverable Plot 4.1:** Dynamic shedding envelope diagram: plot oscillation amplitude $\Delta C_L = C_{L,\max} - C_{L,\min}$ (error whiskers) as a function of degrees past stall $(\alpha - \alpha_{\text{stall}})$.
- **Deliverable Plot 4.2:** Cambered airfoil stall asymmetry comparison: contrast positive vs inverted stall modes.

4 Classroom Consortium: Multi-Variable Data Pooling

Following data acquisition, each division will deposit their calibrated data points into the **Consortium Master Telemetry Database**. Every team will then have access to the complete 4D matrix.

Div.	Airfoil / Case	m	t/c	Re	Meas. α_{stall} (°)	Meas. $C_{L,\max}$
1	NACA 0006	0.00	0.06	1000	_____	_____
1	NACA 0009	0.00	0.09	1000	_____	_____
1	NACA 0012	0.00	0.12	1000	_____	_____
1	NACA 0018	0.00	0.18	1000	_____	_____
1	NACA 0024	0.00	0.24	1000	_____	_____
2	NACA 1412	0.01	0.12	1000	_____	_____
2	NACA 2412	0.02	0.12	1000	_____	_____
2	NACA 4412	0.04	0.12	1000	_____	_____
2	NACA 6412	0.06	0.12	1000	_____	_____
3	NACA 2412 ($Re = 300$)	0.02	0.12	300	_____	_____
3	NACA 2412 ($Re = 600$)	0.02	0.12	600	_____	_____
3	NACA 2412 ($Re = 1200$)	0.02	0.12	1200	_____	_____
3	NACA 2412 ($Re = 2500$)	0.02	0.12	2500	_____	_____
3	NACA 2412 ($Re = 5000$)	0.02	0.12	5000	_____	_____
4	NACA 4406	0.04	0.06	1000	_____	_____
4	NACA 4418	0.04	0.18	1000	_____	_____
4	NACA 2412 (Inverted)	0.02	0.12	1000	_____	_____

Table 1: Consortium Master Parametric Telemetry Table for class-wide regression.

5 Analysis & Discussion Prompts

Each research division will author a technical report synthesizing their primary domain and the global consortium dataset. Address the following analytical prompts:

1. **Thickness Scaling & Optimal Slenderness:** Plot α_{stall} against (t/c) for the symmetric NACA 00XX series.
   - At what thickness ratio does α_{stall} peak?
   - Physically explain the stall mechanism transition between NACA 0006 (thin-airfoil bubble bursting) and NACA 0018 (progressive trailing-edge stall).
   - Why does increasing thickness beyond $t/c \approx 0.18$ eventually degrade aerodynamic performance?
2. **Camber Effect on Stall vs Lift Capability:** Contrast the effect of camber m on $C_{L,\max}$ versus its effect on α_{stall}.
   - Does adding camber increase or decrease the stall angle α_{stall}?
   - Why does NACA 6412 achieve higher $C_{L,\max}$ than NACA 0012 despite intense suction peaks?
   - Verify Thin Airfoil Theory: calculate the linear regression slope $dC_L/d\alpha$ and compare to 2π. Calculate the zero-lift angle α_0 and compare to theoretical $-1.1m$.

3. **Reynolds Scaling and Boundary Layer Energization:** Examine the five Reynolds number test points.
   - Fit an empirical logarithmic model: $\alpha_{\text{stall}} = A + B\ln(Re/1000)$. Report constants A and B and R^2.
   - Relate your findings to the laminar boundary layer thickness $\delta \propto 1/\sqrt{Re}$. Why does an energized, thinner boundary layer delay separation?
4. **Post-Stall Dynamics & Force Vector Behavior:** Observe the real-time vector overlay $(\vec{L}, \vec{D}, \vec{R})$ and smoke streamlines across stall.
   - How does the orientation of the resultant force vector $\vec{R}$ rotate as the airfoil transitions into deep stall?
   - In deep stall, how does the shedding oscillation envelope $[C_{L,\min}, C_{L,\max}]$ relate to the formation and shedding of von Kármán vortex street structures in the wake?
5. **Global Multi-Variable Regression Model:** Using the pooled class dataset from Table 1, fit a multi-linear response surface:

$$\alpha_{\text{stall}}\left(m,\, \frac{t}{c},\, Re\right) \approx C_0 + C_m \cdot m + C_t \cdot \left(\frac{t}{c}\right) + C_{Re} \cdot \ln\left(\frac{Re}{1000}\right). \tag{8}$$

   Determine the regression coefficients C_0, C_m, C_t, C_{Re} and discuss the dominant sensitivity governing stall onset in subsonic flight.

6 Grading Rubric (100 Points Total)

Criterion	Performance Expectations	Points
Experimental Protocol & DAQ Rigor	Correct execution of 5s burst integration protocol; flow settled state verified; appropriate angle density (0.5° near stall); complete CSV exports.	20
Data Visualization & Polar Curves	Clear, publication-grade polar plots (C_L vs α, C_D vs α, L/D vs α) with error whiskers, stall points clearly identified, theoretical references included.	20
Physical Mechanisms & Fluid Dynamics	Sound, thorough physical reasoning explaining separation, suction peaks, boundary layer energy, vortex shedding, and thin vs trailing-edge stall.	25
Consortium Synthesis & Regression	Successful integration with class dataset; formulation and statistical evaluation of the multi-variable scaling model $\alpha_{\text{stall}} = \Phi(m, t/c, Re)$.	20
Technical Communication & Formatting	Clear engineering report organization; proper unit annotations; professional tables, diagrams, and executive summary.	15
Total		**100**

Table 2: Laboratory Evaluation and Assessment Rubric.